

a

# 1 Per-layer bytes

**int4 KV** = 528 B / token /  
GQA layer

fp32 / reference

**fp32 temporal + bf16  
conv**  
= 1,085,440 B / GDN layer

**bf16**

**bf16 temporal + bf16  
conv**  
= 561,152 B / GDN layer

**recurrent bytes**  
-48.30%  
(not exactly half)

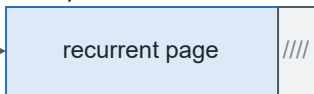
# 2 Padded recurrent page

**B = padded recurrent-page  
bytes / 528 B per token**

fp32 / reference

1,089,792 B

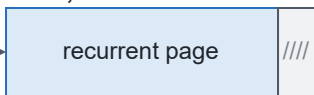
**B = 2,064 tokens**



**bf16**

566,016 B

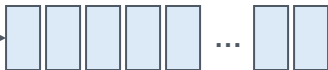
**B = 1,072 tokens**



# 3 Shared fixed GPU pool

fp32 / reference

**K = 3,287 allocated blocks**



**bf16**

**K = 6,330 allocated blocks**



# 4 Discrete capacity at L = 4,096

$$S_{\text{alloc}} = K / (H + \text{ceil}(L/B)), H = 3$$

fp32 / reference

$$3,287 / (3 + 2) = 657.4$$

allocator-equivalent slots

**bf16**

$$6,330 / (3 + 4) = 904.3$$

allocator-equivalent slots

**These are allocator-equivalent slots,  
not concurrent requests.**

*No scheduler behavior, request admission, throughput, or latency is implied.*